

EQUALISATION IN HIGH-SPEED SERIAL LINKS

From a measured channel to a closed link budget — characterisation, transmit and receive equalisation, and the road to 224 Gb/s

September 2026 • Prepared for Brendan Lynskey

What this report is about, in plain terms. A serial link is a filter you did not design and cannot change, followed by a filter you did design and can. This report takes one concrete channel — 28.8 inches of differential stripline across two line cards and a backplane, carrying 28 GBd — and works it all the way through: what the S-parameters say, what the pulse response says, why the eye is completely shut before any equalisation, what each equaliser block can and cannot fix, and what the noise budget looks like once they are all in place. The answer, for this channel, is that the link lands 1.36 dB short of a 10^{-12} error rate, so the report also works through which of the available remedies actually recovers that margin — and which, despite being the obvious thing to reach for, does not. It is a companion to 'Matrix Methods in Network Parameters' and 'Matrix Concepts in Digital Filter Design': the channel is described by the mixed-mode S-parameters of the first, and every equaliser in it is an instance of the optimal-tap problem of the second.

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The channel, in one line. 28 GBd NRZ, so a unit interval of 35.7 ps and a Nyquist frequency of 14 GHz. 28.8 inches of 100 Ω differential stripline on Dk 3.7 laminate at 163 ps per inch, split across a transmit package, a 6-inch line card, an 18-inch backplane, a 4-inch line card and a receive package, with two backplane connectors and the vias that go with them.

1 What closes an eye

A transmitter launches a clean rectangular symbol and a receiver, twenty-nine inches later, sees something that barely resembles it. Four mechanisms are responsible, and it is worth separating them because each is fixed by a different block.

The first and largest is **frequency-dependent loss**. Copper loss rises as the square root of frequency because current crowds into an ever thinner skin, and dielectric loss rises roughly linearly because the laminate's molecules absorb more energy the faster you ask them to reorient. For the laminate used here the two together come to 0.83 dB per inch at 14 GHz. A square pulse is a wide band of frequencies, so a channel that attenuates the high ones and passes the low ones does not merely shrink the pulse — it smears it across many symbol periods. That smearing is **intersymbol interference**, and it is the dominant impairment in every channel in this report.

The second is **reflection**. Every impedance discontinuity — a via, a connector footprint, a package ball — sends part of the wave back, and part of that returns again from the next discontinuity. The result is energy arriving late, which is ISI with a longer memory and a less predictable shape than loss-induced smearing. The third is **crosstalk** from neighbouring lanes, which unlike the first two is not correlated with your own data and therefore behaves as noise rather than as ISI. The fourth is **jitter**, which moves the sampling instant rather than the level, and which converts into amplitude error in proportion to the slope of the signal at the moment you sample it.

Why not just send less far, or use better laminate? Both help, and \$8 prices them. But the economics of a backplane system are set by the chassis, the connector and the laminate cost, all of which are decided long before the SerDes is chosen, and none of which scale with each doubling of data rate. Equalisation is the part of the problem that lives on silicon, which is the only part that gets cheaper every generation. That is why every rate increase for twenty years has been paid for with more equalisation rather than with better copper.

2 A short history

The first serial links worth the name ran at a few gigabits per second over channels that were, by later standards, almost transparent. XAUI in 2002 ran 3.125 GBd over a channel with perhaps 6 dB of loss at Nyquist, and a fixed two-tap transmit de-emphasis — implemented as a switchable current in the output driver — was enough. There was nothing adaptive about it; a board designer chose a setting from a table.

The step that made modern links possible was 10GBASE-KR in 2007, which standardised three things at once: a three-tap transmit feed-forward equaliser, a receive decision-feedback equaliser, and — most importantly — a **training protocol** by which the receiver tells the transmitter which way to move its taps. Until then equalisation had been open-loop and set by hand. After it, a link could negotiate its own settings at start-up, and channels with 20 dB of loss became routine.

CEI-28G, from around 2011, added the continuous-time linear equaliser as a standard receive block and pushed decision feedback from one or two taps to a dozen or more. This is the generation the worked example in this report belongs to, and it is the last one in which a designer can reasonably follow every block by hand. What came next changed the character of the problem: 400 Gigabit Ethernet and CEI-56G, standardised from 2017, moved from two-level signalling to four-level PAM4 and made forward error correction mandatory rather than optional. The 9.5 dB that PAM4 gives away (\$9) is simply not recoverable by equalisation, so the industry stopped trying to reach 10^{-12} raw and started budgeting to a pre-correction error rate of a few times 10^{-4} instead.

IEEE 802.3ck and CEI-112G, from 2022, completed the transition by replacing the analogue slicer with an analogue-to-digital converter and doing the equalisation in the digital domain. Once the waveform is in a register file, the number of taps stops being a question of silicon area per tap and becomes a question of digital power, and algorithms that were impractical in

analogue — long feed-forward filters, maximum-likelihood sequence estimation — become available. The 224 Gb/s generation now being standardised as 802.3dj continues in that direction.

Each generation added a block, not a better cable

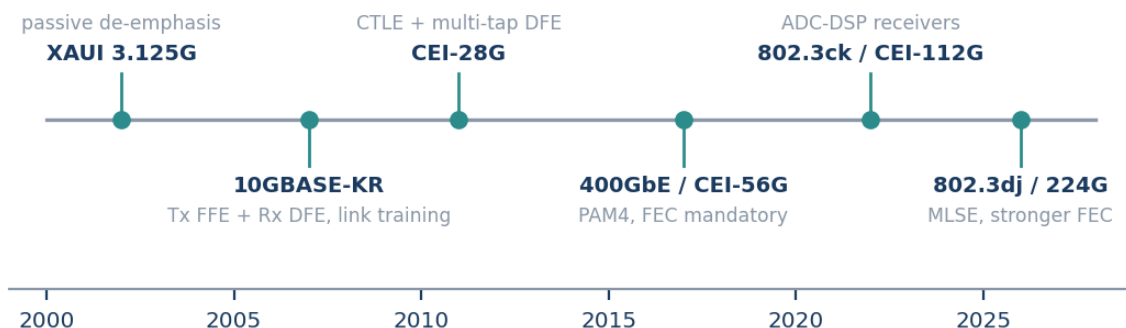


Figure 1 — Twenty-four years of Ethernet and OIF serial links. Each generation bought its rate increase with more equalisation on silicon, not with better copper.

2.1 The same problem, solved again on every other bus

Ethernet and the OIF agreements are only one branch of the family. PCI Express is the other one most engineers meet, and it is worth tracing because it faces the same physics under quite different commercial constraints. Where a backplane Ethernet link is a point-to-point connection between two devices chosen by the same system architect, a PCIe link crosses a socket, a connector and an add-in card from a different vendor, and has to interoperate with parts designed a decade apart. It cannot simply let both ends adapt freely and hope.

Its answer, introduced with PCIe 3.0 in 2010, is a negotiated **preset** table. Rather than requesting arbitrary tap movements, the receiver asks the far-end transmitter for one of eleven predefined combinations of de-emphasis and pre-shoot, and the link equalisation procedure walks through four phases to agree on them. It is a more constrained mechanism than the continuous tap negotiation of 10GBASE-KR, and deliberately so: a finite, published set of transmitter behaviours is what makes cross-vendor interoperability testable.

Generation	Rate per lane	Year	What was added
PCIe 1.0 / 2.0	2.5 and 5.0 GT/s	2003, 2007	8b/10b coding; fixed de-emphasis chosen by the platform
PCIe 3.0	8 GT/s	2010	128b/130b coding, transmitter presets and the four-phase link equalisation procedure — the point at which PCIe became an equalised link
PCIe 4.0 / 5.0	16 and 32 GT/s	2017, 2019	Tighter channel budgets, retimers become common, and connector and card loss start to dominate the budget
PCIe 6.0	64 GT/s	2022	PAM4, FLIT-based framing, and a deliberately lightweight forward error correction backed by link-level retry
PCIe 7.0	128 GT/s	2025 (draft)	PAM4 at 64 GBd; the reach question becomes acute and optical options enter the specification discussion

Why PCIe chose a weaker code than Ethernet. Both moved to PAM4 and both had to add forward error correction to survive it, but they made opposite trades. Ethernet tolerates the hundred nanoseconds or so that a Reed-Solomon KP4 decoder costs, because a hundred nanoseconds is nothing beside the latency of the network it sits in. PCIe is a load-store fabric where a processor may be stalled waiting for the answer, so latency is part of the contract. PCIe 6.0 therefore pairs a much lighter FEC — one that corrects only a modest number of symbols — with a cyclic redundancy check and a link-level retry for whatever the FEC misses. Same physics, same constellation, different answer, because the cost function is different.

The pattern repeats across the other serial buses, and the interesting axis to compare them on is reach rather than rate. USB grew from 5 Gb/s in 2008 to 20 Gb/s per lane with USB4, and then to 80 Gb/s in USB4 version 2 by adopting three-level PAM3 signalling — a compromise that buys throughput with less of a signal-to-noise penalty than PAM4 while still needing careful equalisation over a consumer-grade cable. SAS reached 22.5 Gb/s, InfiniBand went from 25 Gb/s NRZ at EDR to 50 and then 100 Gb/s per lane in PAM4 at HDR and NDR, and DisplayPort and HDMI carry comparable rates over cables with no ground reference to speak of.

At the far end of the axis sits UCle, the die-to-die interface standardised from 2022. Its channel is a couple of millimetres of silicon interposer rather than tens of inches of laminate, and at that reach the loss is negligible, the equalisation is nearly nothing, and the design problem becomes one of parallelism and power per bit. That is the direction every one of these standards is being pushed: as the rate climbs, the reach that equalisation can rescue falls, and the answer is either to shorten the copper or to stop using it.

3 The channel: layout, and what the S-parameters say

3.1 The physical channel

The channel is a conventional two-card backplane topology. A transmitter drives out through its package escape, across six inches of line-card stripline, down through a via into a backplane connector, along eighteen inches of backplane, up through a second connector, across four inches on the receiving line card, and into the receiver's package. Twenty-eight point eight inches of copper in total, on a laminate with a dielectric constant of 3.7, which gives 163 picoseconds of delay per inch and a total flight time of 4.7 nanoseconds — a hundred and thirty-one unit intervals. There are always more than a hundred symbols in flight at once, which is worth remembering when reasoning about what a reflection does.

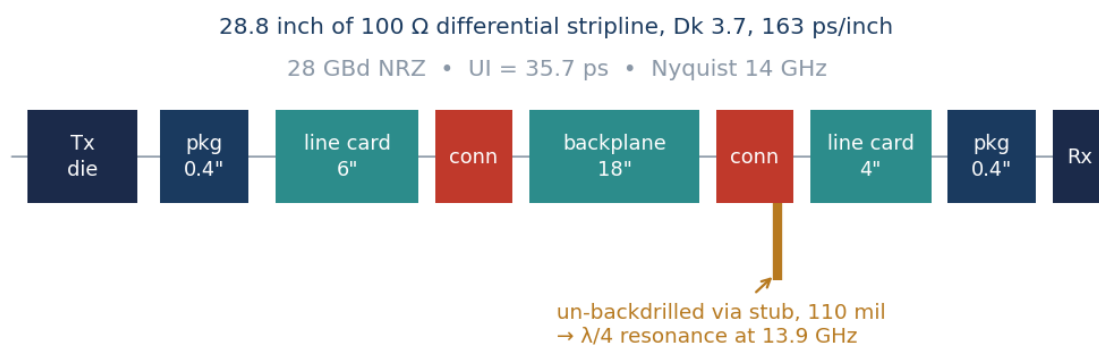


Figure 2 — The channel. Two connectors, four via transitions and one of them left un-backdrilled.

3.2 What the laminate is, and why it matters more each generation

A dielectric constant of 3.7 and a loss tangent of 0.0079 are not arbitrary numbers: they identify a class of material. Dielectric loss is, to a good approximation, $\alpha_d \approx 2.3 f \sqrt{Dk} \cdot Df$ decibels per inch with f in gigahertz, so the loss tangent Df sets the slope of the channel directly while the dielectric constant Dk sets the propagation delay and, through it, how much copper you need for a given reach. The baseline here is a mid-loss laminate of the kind that was standard for 10 Gb/s work and is now marginal.

Laminate	Dk	Df	Dielectric loss at 14 GHz	Typically used for
Standard FR-4	≈4.3	≈0.022	1.47 dB/inch	Up to a few Gb/s — PCIe 1.0 and 2.0, USB 3.0, XAUI
Isola FR408HR	3.65	0.0092	0.57 dB/inch	8 to 10 Gb/s — PCIe 3.0, 10GBASE-KR
This report's channel	3.70	0.0079	0.49 dB/inch	Mid-loss; adequate at 10 Gb/s, marginal at 28
Isola I-Speed	3.53	0.0060	0.36 dB/inch	16 to 28 Gb/s — PCIe 4.0 and 5.0, CEI-28G
Panasonic Megtron 6	3.55	0.0045	0.27 dB/inch	25 to 28 Gb/s backplanes; the low-loss remedy of §8
Rogers RO4350B	3.48	0.0037	0.22 dB/inch	RF and mixed-signal boards where Dk stability matters most
Isola Tachyon 100G	3.02	0.0021	0.12 dB/inch	56 Gb/s PAM4 and above — CEI-56G, PCIe 6.0
Panasonic Megtron 7	3.35	0.0020	0.12 dB/inch	112 Gb/s per lane — 802.3ck, CEI-112G

Two things that do not appear in a datasheet Dk and Df often matter as much as the resin does. The first is **copper roughness**. A foil is deliberately roughened so that it adheres to the laminate, and at microwave frequencies the current follows that roughened surface rather than cutting across it, lengthening the path and raising the conductor loss. Standard foil with a profile of five to seven micrometres can add half again to the smooth-conductor loss at 14 GHz; reverse-treated foil brings that down, very-low-profile foil further, and the hyper-VLP grades used above 50 GBd further still. A board built on excellent resin with ordinary foil can easily lose to one built on worse resin with better copper.

The second is the **glass weave**. Laminate is resin reinforced with woven glass cloth, and glass and resin have different dielectric constants — roughly 6 against 3. A differential pair whose two conductors happen to run one over the glass bundles and the other over the resin therefore sees two different propagation velocities, which is intra-pair skew, which is mode conversion by §9 of the companion report. It accumulates with length and it is the reason long pairs are routed at a small angle to the weave, or on the flattened, spread-glass styles such as 1078 and 3313 that even out the difference.

Why the RF laminates are not the answer. The one genuinely surprising row in the table is Rogers RO4350B, which at 0.22 dB per inch is lower loss than the Megtron 6 that §8 recommends — and yet nobody builds a backplane out of it. The reasons are all about system fit rather than decibels. It is engineered for Dk stability, ± 0.05 over temperature and lot, which matters when the dielectric constant sets a filter passband or an antenna length and is of no use whatever to a serial link. Its Dk of 3.48 is also too high: digital laminates chase low Dk deliberately, because a lower dielectric constant allows wider traces at the same 100 Ω differential impedance, and conductor loss falls roughly as trace width rises. And the conductor term is not a detail at these frequencies — it is comparable to the dielectric term, and it is dominated by foil roughness, where differences of 0.1 to 0.2 dB per inch between grades are typical. The 0.05 dB per inch that RO4350B gains on Megtron 6 in the dielectric is smaller than the foil advantage it does not come packaged with, because the digital vendors sell core, prepreg, glass style and foil profile as one qualified system and the RF vendors offer a narrower range.

Two more practical objections finish the case. RF laminates are not offered in the spread-glass constructions that control intra-pair skew, because a single-ended RF line does not care about weave and a differential pair very much does. And the bondply system and the reliability qualification behind a material such as RO4350B target thin, low-layer-count boards, not a forty-layer backplane built by sequential lamination with tens of thousands of plated through-holes to survive thermal cycling — on top of which the ceramic filler is abrasive enough to matter to drill life, and the laminate costs several times as much per square foot over a large panel. Where these materials do appear in a high-speed digital product is as an RF island hybrid-laminated into an otherwise digital stackup, which is the sensible way to use them. The PTFE-based laminates take every one of these trade-offs further in the wrong direction: lower loss again, but poorer dimensional stability, special processing for adhesion, and worse through-hole reliability in a thick board.

How the generations map onto materials. Up to a few gigabits per second, ordinary FR-4 and ordinary foil. At 8 to 10 Gb/s, an improved FR-4 with reverse-treated foil. At 16 to 28 Gb/s, a genuine low-loss laminate, very-low-profile copper and spread glass. At 56 and 112 Gb/s per lane, an ultra-low-loss resin with hyper-VLP foil — and a much shorter channel, because at those rates no material saves a backplane. That last clause is the important one: materials bought perhaps 15 dB of improvement across twenty years, while the loss at Nyquist of a fixed length of copper roughly tripled over the same period.

3.3 Insertion loss, and a via stub in exactly the wrong place

The differential insertion loss S_{dd21} is the single number a link budget starts from. Here it is -4.0 dB at 1 GHz and -33.3 dB at the 14 GHz Nyquist frequency — a hard channel, at the limit of what 28 GBd NRZ is normally asked to cross, but not an unreasonable one.

That figure assumes the vias have been back-drilled. If one of them is left with the unused barrel intact, the result is a 110 mil open-circuited stub hanging off the through path, and an open stub is a quarter-wave short at the frequency where its length equals a quarter wavelength. At 163 ps per inch that lands at 13.9 GHz, which is within a percent of Nyquist. The stub adds 25 dB of loss at exactly the frequency the link can least afford it, taking the channel from -33.3 dB to -58.2 dB. No amount of equalisation recovers that. Back-drilling is not an optimisation; on this channel it is the difference between a link and a brick.

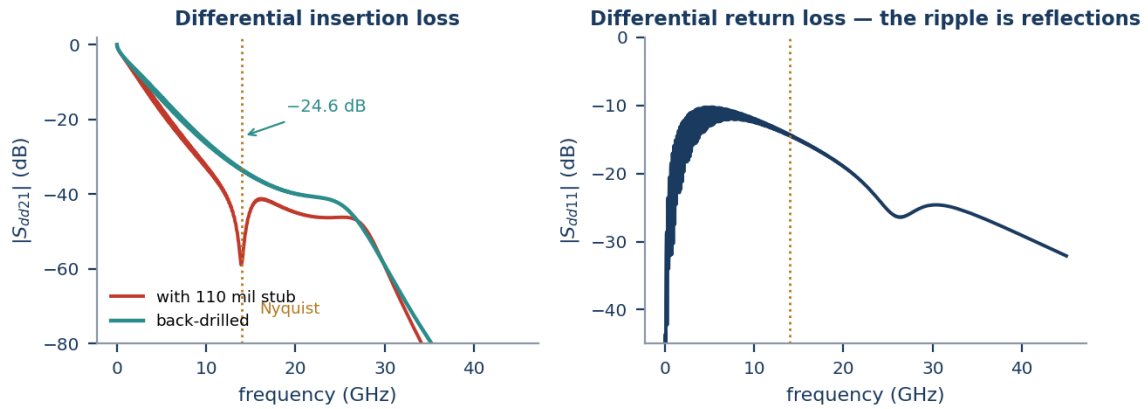


Figure 3 — Differential insertion loss with and without the via stub, and the differential return loss. The stub notch sits within a percent of Nyquist.

3.4 Return loss, crosstalk and mode conversion

Return loss matters less directly than insertion loss but is not ignorable. Here S_{dd11} peaks at about -15 dB near Nyquist, and the ripple visible on the insertion-loss curve is the signature of energy bouncing between the two connector transitions. Each bounce is a delayed copy of the signal, and because the round trip between connectors is many unit intervals long, those copies land as isolated ISI terms far from the cursor — exactly where a decision-feedback equaliser with a handful of taps cannot reach them.

Crosstalk does not appear in a two-port measurement at all, which is one reason characterisation is done on a multiport fixture. Near-end crosstalk from lanes driving in the opposite direction and far-end crosstalk from lanes driving alongside are aggregated into a single figure of merit, the integrated crosstalk noise, which is what enters the budget in §7 as 3.0 mV rms. And mode conversion, the S_{cd} block of the companion report, matters here for a reason specific to equalisation: converted energy is energy the receiver never sees, so it reduces the cursor without appearing anywhere in the differential insertion loss.

4 From S-parameters to a pulse response

Insertion loss tells you how much of a sine wave survives. It does not directly tell you what a receiver sees, because a receiver does not sample sine waves — it samples one symbol at a time, in the presence of all the others. The object that answers that question is the **pulse response**: the voltage waveform produced by sending a single one-unit-interval symbol and nothing else. Take the inverse Fourier transform of S_{dd21} to get the impulse response, convolve with a one-UI rectangle, and you have it.

Sample that waveform once per unit interval, at the phase that maximises the peak, and the resulting sequence of numbers is the whole story. The largest sample is the **cursor**: the part of the symbol that arrives when it is supposed to. Samples before it are **pre-cursors**, energy that has arrived early; samples after it are **post-cursors**, energy still arriving one, two, ten symbols later. Every one of them lands on top of a neighbouring symbol.

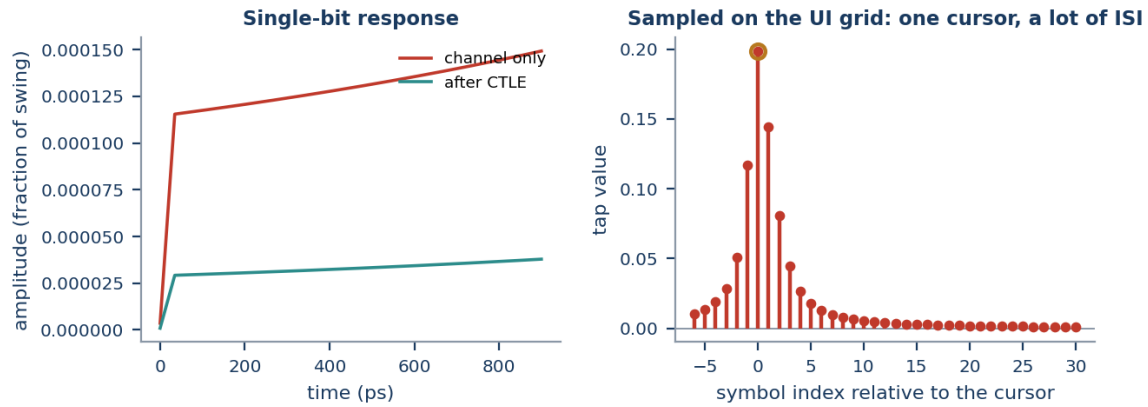


Figure 4 — The single-bit response before and after the CTLE, and the same waveform sampled on the unit-interval grid. The circled sample is the cursor; everything else is interference.

The closed eye, in numbers. For this channel the cursor is 0.199 of the transmitted amplitude, and the sum of the absolute values of every other tap is 0.636 — nearly three times as large. The worst-case eye opening is twice the difference between the two, which comes out at -0.87. A negative eye opening is not a small eye; it means that for some patterns of neighbouring data the receiver sees the wrong sign entirely. The first post-cursor alone is 0.144 and the first pre-cursor 0.117, each comparable with the cursor itself. Nothing about this link works without equalisation.

5 The equalisers, one block at a time

5.1 The continuous-time linear equaliser

The CTLE is an analogue filter in the receiver front end, usually a degenerated differential pair whose source degeneration is bypassed by a capacitor so that gain rises with frequency. It is the cheapest equalisation there is: a handful of devices, no clock, no adaptation loop beyond selecting one of a dozen settings. What it does is approximately invert the channel's slope over the band of interest. What it cannot do is add gain — a CTLE with a boost of 8.3 dB achieves it by attenuating low frequencies, not by amplifying high ones, so the cursor shrinks along with the ISI. That is less damaging than it sounds, because the front-end noise and the crosstalk arriving alongside the signal are attenuated by the same filter: over the receiver's noise bandwidth this setting has a gain of -5.6 dB, so what a CTLE really trades is signal against ISI rather than signal against noise.

That is exactly what happens here. The CTLE brings the ISI sum down from 0.636 to 0.139, a real improvement, but the cursor falls from 0.199 to 0.070 at the same time. The eye opening improves from -0.87 to -0.14 — still negative, still shut. The CTLE has done useful work and has not, on its own, produced a link. The setting used here is the one a receiver's adaptation converges on — the best of the sixteen or so a real part offers, found by sweeping them and keeping whichever gives the highest signal-to-noise ratio at the slicer.

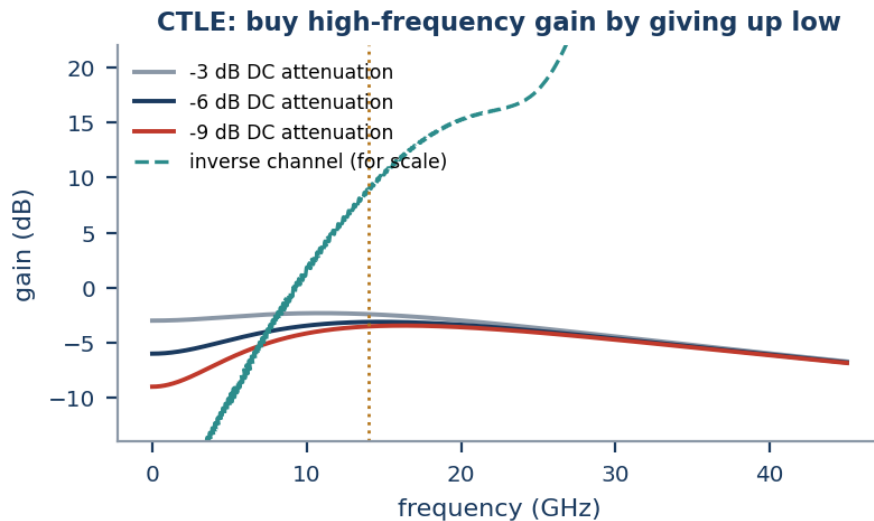


Figure 5 — The CTLE family. More boost is bought by giving away more low-frequency gain. The dashed curve is the inverse channel, for scale: a single pole-zero pair cannot track it far.

5.2 The transmit feed-forward equaliser

The transmitter can pre-distort. A feed-forward equaliser is a short FIR filter running at the symbol rate, and because it operates before the channel it shapes the signal without amplifying anything the channel or the neighbours add afterwards. That is its unique advantage, and it is the reason transmit equalisation survives even though receivers have become far more capable.

Its unique disadvantage is the peak-power constraint. A transmitter has a maximum output swing, so the tap magnitudes must sum to no more than one; every unit of correction is subtracted from the main cursor. The three-tap filter designed here, with two pre-taps set to zero-force the first two pre-cursors, comes out as $[0.040, -0.340, 0.620]$, and the price is visible immediately: the cursor falls again, from 0.070 to 0.032. The eye opening finally turns positive, at -0.007, but only just.

Why the transmitter is asked to handle pre-cursors specifically. A decision-feedback equaliser works by subtracting the known contribution of symbols it has already decided. It therefore cannot touch pre-cursor ISI, which is caused by symbols it has not decided yet. A receive feed-forward filter can, at the cost of noise enhancement. A transmit FFE can, at the cost of swing. Splitting the work so that the transmitter handles the pre-cursors and the feedback path handles the post-cursors is not arbitrary — it puts each impairment where the cheapest remedy is.

5.3 The receive feed-forward equaliser: two effects, not one

The receiver can filter too, and a symbol-rate FIR there is the most flexible linear block available: it sees the signal after the channel has done its worst, it can have many taps, and in a digital receiver it adapts continuously. It is the block that distinguishes a modern ADC-DSP receiver from the analogue one this worked example is built around, and \$8 prices exactly what adding it would be worth here.

It is tempting — and wrong — to describe such a filter as a device that removes interference at the cost of amplifying noise. It does two things at once, and they pull in opposite directions. Consider what happens when the filter forms a weighted sum of several successive samples of the received waveform. The **signal** contributions add coherently: the pre- and post-cursor samples are not noise, they are the same transmitted symbol arriving at the wrong time, so when the filter lines them up and adds them their amplitudes add directly. The **noise**

contributions, being uncorrelated from sample to sample, add only in root-sum-square. Combine k samples carrying comparable signal and the signal grows roughly as k while the noise grows as $\sqrt{k}$, so the ratio improves.

That is the same mechanism as a matched filter, and it has the same limit. The most any receiver can extract from a pulse is set by the total energy in that pulse: the **matched-filter bound**, $\text{SNR} = 2E/N_0$, is what you would get from a filter that collects every scrap of the pulse response and worries about nothing else. A linear equaliser cannot beat that bound, and it usually falls short of it, because it is not free to collect energy wherever it likes — it must also drive the interference to zero, and on a channel with deep spectral nulls those two objectives conflict. The shortfall is what the literature means by **noise enhancement**.

What the number in this report actually measures. The 'noise enhancement' quoted here is not the raw gain of the filter. The nine-tap minimum-mean-squared-error design on this channel has a Euclidean tap norm of about 34, which would be an alarming figure if it were a noise penalty — but the same taps raise the cursor by a comparable factor, because they are collecting signal that the channel had scattered across a dozen symbol periods. The meaningful quantity is the ratio of the two, the noise gain divided by the signal gain, and that comes out at 0.75 dB. That is the net cost, already accounting for the coherent recovery. Quoting the tap norm alone would overstate the penalty by more than twenty decibels.

Two consequences follow. The first is that a linear equaliser is worth having even when the channel is benign, because energy collection is a gain and not merely a rearrangement. The second is the one that shapes real receivers: since the penalty is the gap between coherent signal gain and incoherent noise gain, any mechanism that removes interference without touching the noise at all is strictly better. That is precisely what the next block does, and it is why a decision-feedback tap is preferred to a feed-forward tap wherever a decision-feedback tap will do.

5.4 The decision-feedback equaliser

The decision-feedback equaliser is the one block that removes interference without amplifying noise at all. Having decided a symbol, the receiver knows exactly what that symbol will contribute to the next few, and subtracts it. Because the subtraction is driven by a clean two-level decision rather than by the noisy waveform, no noise rides along with it. Where the feed-forward filter of §5.3 pays a gap between coherent signal gain and incoherent noise gain, the feedback path pays nothing: the decision is noiseless by construction.

This is why the classical optimum has the structure it does. Forney's decomposition of the maximum-likelihood receiver puts a whitened matched filter at the front, whose job is exactly the energy collection of §5.3, and follows it with a detector that resolves the remaining causal interference. The minimum-mean-squared-error decision-feedback equaliser is the practical descendant of that structure: a feed-forward section that gathers the pulse energy and leaves a minimum-phase response, and a feedback section that cancels the causal tail of that response for free. It outperforms the best purely linear equaliser on every channel with significant loss, and the margin between them grows with the loss. What stops it reaching the matched-filter bound in practice is not the mathematics but the next paragraph.

The five taps used here are [0.541, 0.133, 0.038, 0.027, 0.026], and they reduce the residual ISI to 0.349 of the cursor. Two caveats come with them. The first is **error propagation**: a wrong decision is subtracted with the wrong sign, which makes the next decision more likely to be wrong. With a tap as large as 0.54 that is not a remote possibility, and it is one reason forward error correction and decision feedback are usually specified together. The second is that the feedback path must settle within one unit interval — 35.7 ps here — which puts the first tap on the critical timing path of the whole receiver and is why it is often unrolled into a speculative, parallel form.

Block	Removes	Cannot remove	Costs
CTLE	Broadband slope, cheaply	Anything with structure — reflections, isolated echoes	Cursor amplitude; it attenuates rather than amplifies
Transmit FFE	Pre-cursor and near post-cursor ISI, before crosstalk is added	Anything it cannot see; it has no knowledge of the receiver's noise	Transmit swing, through the peak-power constraint
Receive FFE	Pre-cursor ISI and long tails, with many taps available	Nothing in principle, but each tap costs noise	Noise enhancement — here 0.75 dB
DFE	Post-cursor ISI exactly, with no noise penalty	Pre-cursor ISI, ever; it acts only on decided symbols	Error propagation, and a hard timing path on the first tap

6 The worked design, stage by stage

Putting the blocks in series and tracking the pulse response through them gives the table below. It is set out twice on purpose. The left-hand group is in millivolts at the receiver input, which is where a budget has to be written; the right-hand group normalises each stage to its own cursor, which is where the shape of the response is visible without the scaling getting in the way.

Stage	Cursor	ISI (rms)	Non-ISI noise	ISI ÷ cursor	Eye ÷ cursor
Channel only	79.49 mV	88.18 mV	3.50 mV	3.199	-4.397
+ CTLE	27.85 mV	21.33 mV	1.84 mV	1.995	-1.989
+ transmit FFE	12.94 mV	7.34 mV	1.84 mV	1.114	-0.228
+ DFE, five taps	12.94 mV	1.12 mV	1.84 mV	0.349	1.303

6.1 Which blocks must shrink the cursor, and which need not

The absolute column falls sharply, and it is worth being precise about why, because the four blocks are not alike and two of them are shrinking the cursor for quite different reasons.

Block	Does it scale the cursor?	Why
CTLE	Yes, and it must	It is a passive-shaped analogue filter with a fixed headroom. Its high-frequency gain is about unity, so the only way it can present a rising response is to attenuate everything below. But it attenuates the incoming noise and crosstalk by nearly the same factor — here -5.6 dB over the noise bandwidth — so most of the cursor reduction is not a loss of signal-to-noise ratio
Transmit FFE	Yes, and it must	The transmitter has a hard peak-power limit, so $\sum c_i \leq 1$ and every unit of correction is subtracted from the main tap. The noise it fights is added downstream and does not scale with it , so this attenuation is a real signal-to-noise loss — the most expensive decibels in the chain

Block	Does it scale the cursor?	Why
Receive FFE	No, not meaningfully	Its gain is arbitrary because a variable-gain amplifier and the slicer threshold follow it and scale together. Only the ratio of its signal gain to its noise gain matters (§5.3), which is why it is fair to normalise its output to unity
DFE	No	It subtracts rather than scales. The cursor passes through untouched and the noise is untouched too, because the thing being subtracted is a decision rather than a waveform

6.2 What actually falls monotonically

It is often said that equalisation works by making the interference shrink faster than the signal does. That is a statement about the cursor, and the cursor is not a well-defined quantity: it depends on where you put the gain, and slide the automatic gain control and you can make it whatever you like. The invariant statement is about ratios, and it is this:

Every linear block in the chain trades signal-to-noise ratio for signal-to-ISI ratio. Measured against the noise that is not correlated with the data — thermal noise in the front end, crosstalk from the neighbours, quantisation in the converter, jitter converted to amplitude through the slope — the signal falls monotonically through the chain, from 27.1 dB at the channel output to 16.9 dB after the transmit FFE, and no linear operation can ever reverse that. What is bought in exchange is the other ratio: signal to residual ISI climbs from -0.9 dB to 21.3 dB. The link works when the two together are good enough, and both matter only over the bandwidth the signal actually occupies.

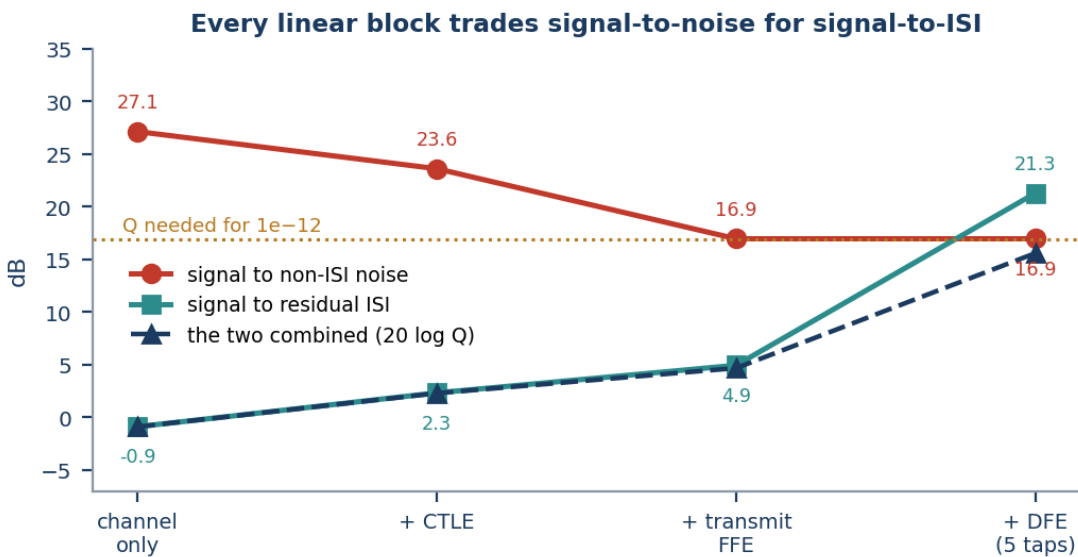


Figure 6 — The scale-invariant picture. Signal to non-ISI noise (red) can only fall; signal to residual ISI (green) is what the equalisers buy. The decision-feedback stage is the only one that improves the second without spending any of the first.

The figure also shows why the division of labour of §5 is not merely tidy. The decision-feedback equaliser is the only block whose signal-to-noise line is flat: it improves the ISI ratio by 16.3 dB at no cost in signal-to-noise ratio whatsoever. Every other block pays. And the transmit FFE, judged on its own, looks like a poor bargain — it gives up 6.7 dB of signal-to-noise to buy only 2.6 dB of signal-to-ISI. Run the chain without it and the five feedback taps take Q only from 1.30 to 1.69, because what remains after the CTLE is dominated by pre-cursors that the feedback path cannot reach. The transmit FFE is not there

to improve the eye by itself; it is there to clear the pre-cursors so that the block which is free can do its work. Together they take Q from 1.30 to 6.01.

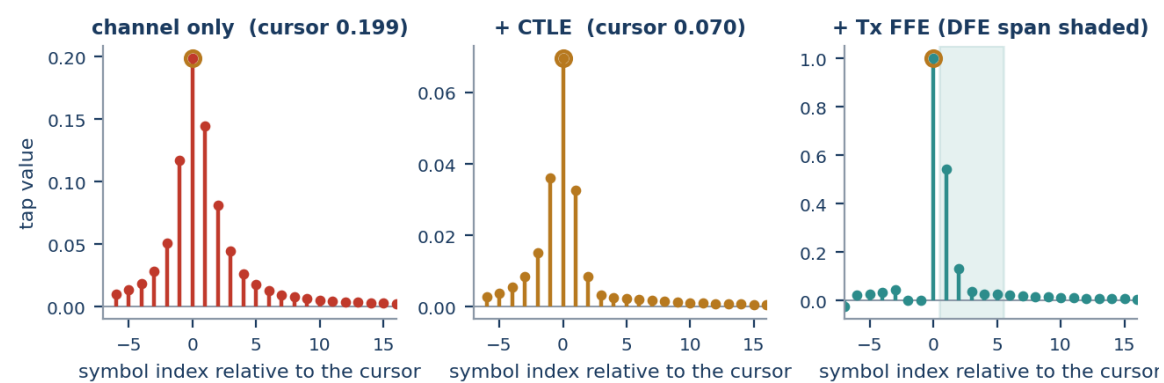


Figure 7 — The sampled pulse response at each stage. Note the vertical scales: the CTLE cuts the cursor to about a third while cutting the ISI sum by nearly five, which is why it is worth having. The final panel is normalised to its own cursor, with the five feedback positions shaded.

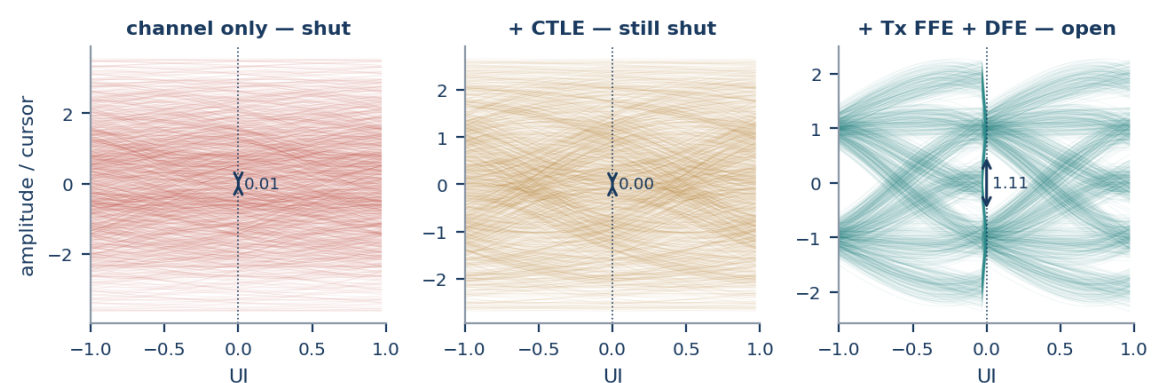


Figure 8 — The same three stages as eye diagrams, built by driving four thousand random symbols through the modelled chain. The arrow marks the eye opening at the sampling instant in units of the cursor amplitude, so an ideal ISI-free eye would read 2.0 and the equalised eye at 1.11 is a little over half of that.

7 The noise and jitter budget

An open eye is necessary but not sufficient. What decides the error rate is the ratio of the eye opening to everything that moves the received level around, and those contributions have to be referred to a common point. Referring them to the receiver input, where the cursor is 12.9 mV out of a 800 mV transmitted swing, gives the budget below.

Contribution	Value	Where it comes from
Receiver noise	0.95 mV rms	1.8 mV of front-end noise, shaped by the CTLE, whose gain over the noise bandwidth is -5.6 dB
Crosstalk	1.58 mV rms	3.0 mV of integrated crosstalk noise from neighbouring lanes, shaped by the same filter
Jitter	0.10 mV rms	350 fs rms of random jitter, converted to amplitude through the slope of the waveform at the sampling instant
Residual ISI	1.12 mV rms	What the equalisers left behind, treated statistically rather than worst-case
Total	2.15 mV rms	Root-sum-square of the above

A note on the letter Q, because it is badly overloaded. The Q used here is the Q-factor of digital and optical communications, and it has nothing whatever to do with the quality factor of a resonator that appears in §5 of the companion report. It is named after the Gaussian tail function $Q(x) = \int_x^\infty \phi(u) du$, and it is defined as the separation of the two signal levels divided by the sum of their noise standard deviations, $Q = (\mu_1 - \mu_0)/(\sigma_1 + \sigma_0)$, which for a symmetric binary signal is simply half the eye opening divided by the rms noise. The bit error rate follows as $\frac{1}{2} \operatorname{erfc}(Q/\sqrt{2})$, so $Q = 7.03$ is 10^{-12} and $Q = 6$ is about 10^{-9} . Optical work usually quotes it in decibels as $20 \log Q$, which invites a further confusion with electrical signal-to-noise ratio in power terms.

The collision is worth naming because it genuinely bites anyone working across disciplines. In an atomic clock, for instance, the fractional frequency stability depends on the resonator Q and on the signal-to-noise ratio, and the two appear in the same expression — so an engineer who reads 'Q' in this report as the resonator quantity will get a sentence that is not merely wrong but wrong in a plausible-looking way. Where the ambiguity would matter, the safest course is to quote the error rate directly, or to use the IEEE 802.3 **channel operating margin**, which expresses the same judgement as a single number in decibels and is defined in the standard rather than by convention. This report keeps Q because it is what every SerDes datasheet and every optical link budget uses, but it means the communications Q-factor throughout and never the resonator one.

The cursor is 12.9 mV and the total noise 2.15 mV rms, so the Q-factor is $Q = 6.01$. That corresponds to a bit error rate of about 10^{-9} . The conventional target for an uncorrected link is 10^{-12} , which needs $Q = 7.03$, so this link is **1.36 dB short**.

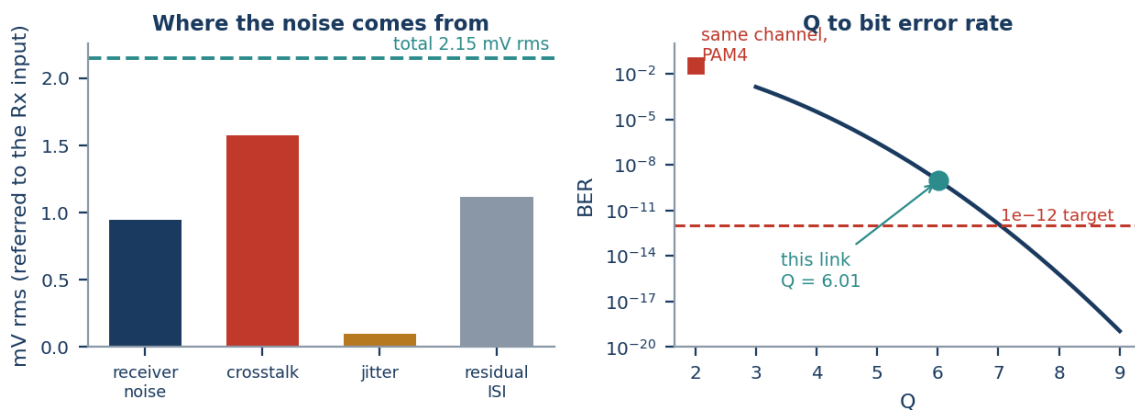


Figure 9 — Where the noise comes from, and what the resulting Q buys. Crosstalk dominates; residual ISI is close behind it, and receiver noise third.

Read the budget before choosing a fix. The largest single contribution is crosstalk at 1.58 mV, with residual ISI second at 1.12 mV and receiver noise third. That ordering matters, because it says the link is short of margin mostly because of its neighbours and only partly because of what the equalisers left behind. Removing every last scrap of residual ISI would take the total from 2.15 mV down to about 1.8 mV, worth roughly four tenths of a decibel — nothing like the shortfall. §8 checks that arithmetic against the alternatives.

8 It does not close: what would actually fix it

Seven candidate remedies, each re-run through the whole chain from the channel model outwards, so that the margin figures below are computed rather than estimated.

What you change	Loss at Nyquist	Q	Margin to 10^{-12}
As built	-33.3 dB	6.01	-1.36 dB
Halve the crosstalk (connector + pair pitch)	-33.3 dB	7.77	+0.87 dB
Low-loss laminate (Df 0.005 → 0.003)	-28.0 dB	8.98	+2.12 dB
Shorten the backplane 18" → 13"	-29.3 dB	8.52	+1.67 dB
Ten DFE taps instead of five	-33.3 dB	6.20	-1.10 dB
Add a 9-tap receive FFE (the ADC-DSP answer)	-33.3 dB	6.34	-0.90 dB
Low-loss laminate + halved crosstalk	-28.0 dB	11.38	+4.18 dB

The table divides cleanly, and the division is the point. Everything that changes the board is worth between one and a half and two and a half decibels. A lower-loss laminate lifts the cursor by reducing the slope the equalisers have to undo; five inches off the backplane does much the same and is often cheaper if the mechanical design still has any freedom in it; halving the crosstalk attacks the largest single term in the budget directly. Any one of the three closes the link on its own.

Everything that changes the silicon is worth a fraction of that. Doubling the decision-feedback taps from five to ten buys 0.26 dB. Adding a nine-tap receive feed-forward equaliser — the block that defines a modern ADC-DSP receiver, and the obvious thing to reach for — buys 0.46 dB, and still leaves the link short of target. Neither is useless, and on a channel dominated by intersymbol interference rather than by crosstalk both would look very different. But on this channel the budget said the problem was the neighbours, and the budget was right.

The general lesson. The block that fixed the last problem is rarely the block that fixes the next one. A link short of margin is short for a reason, the budget names the reason, and the instinct to reach for more equalisation is the instinct to solve the problem you solved last time. Build the budget first: it costs an afternoon, and it tells you which of seven candidate fixes are worth the meeting.

9 The same channel at 56 Gb/s: PAM4 is not free

The obvious way to double throughput without doubling the Nyquist frequency is to send two bits per symbol. PAM4 does exactly that, with four amplitude levels instead of two, at the same 28 GBd and therefore across the same 14 GHz of channel. The channel does not get any worse. The signalling does.

Four levels spread across the same peak-to-peak swing leave three eyes, each one third the height of the single NRZ eye. That is a loss of $20 \log 3 = 9.5$ dB of signal-to-noise ratio before anything else is considered, and it is not recoverable by equalisation because it is a property of the constellation rather than of the channel. On this channel the Q of 6.01 becomes 2.00, and the raw error rate rises from 10^{-9} to about $10^{-1.5}$.

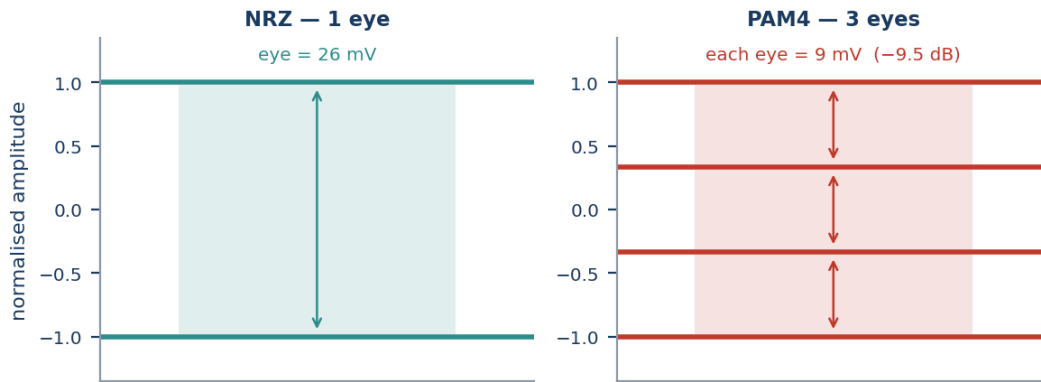


Figure 10 — The same swing, divided two ways. Three eyes of one third the height cost 9.5 dB before the channel is even considered.

An error rate of one in thirty is not a link, which is why PAM4 generations made forward error correction mandatory rather than optional. The Reed-Solomon KP4 code, RS(544,514) over GF(2^{10}), takes a pre-correction error rate of about $2.4\text{e-}04$ down below 10^{-15} , and PAM4 links are budgeted to that threshold instead of to 10^{-12} . Even so, this channel does not reach it: the required Q is 3.60 and the available Q is 2.00, a shortfall of 5.1 dB.

Channel change	Loss at Nyquist	Q	Margin to the KP4 threshold
PAM4, as built	-33.3 dB	2.00	-5.08 dB
PAM4 + low-loss laminate	-28.0 dB	2.99	-1.60 dB
PAM4 + low-loss + 13" backplane	-25.1 dB	3.97	+0.84 dB
PAM4 + all of the above, plus a receive FFE	-25.1 dB	6.30	+4.87 dB

The lesson is the one the industry learned between 2015 and 2020. PAM4 does not let you run twice the data over the channel you already have; it lets you run twice the data over a channel roughly 8 dB better than the one you already have, in exchange for a coding layer, a larger latency budget and a considerably more complicated receiver.

10 Adaptation and training

Nothing in §5 can be set from a datasheet. Manufacturing spread, temperature, the particular board and the particular connector all move the channel, so every tap in a modern link is adapted. The mechanism is the least-mean-squares algorithm of the companion report on digital filters, usually in its sign-sign form: take the sign of the error between the sampled level and the decided level, take the sign of the data in the relevant tap position, multiply, and accumulate. That requires no multiplier and no knowledge of the channel, and it converges to the same minimum of the same quadratic bowl that the Wiener-Hopf equations describe.

Receive-side taps can be adapted continuously because the receiver has the error signal in hand. Transmit-side taps cannot, because the transmitter does not know what the receiver saw, so the standards define a **link training** phase: before data flows, the two ends exchange a control channel over which the receiver requests increment, decrement or hold on each transmit tap, and the transmitter reports what it did. 10GBASE-KR introduced this and every backplane standard since has kept it. Training typically takes a few milliseconds and is re-run on every link-up.

Adaptation interacts with the clock. The clock and data recovery loop chooses the sampling phase, and the equaliser adapts to whatever phase it is given, while the phase detector sees a waveform shaped by whatever the equaliser is currently doing. The two loops can and do fight, and a link that trains to a stable but poor setting — the wrong lock point, an eye centred half a unit interval away from the best one — is a common and deeply unpleasant failure. Bandwidth separation between the loops is the usual defence.

11 Where this is going

The 224 Gb/s per lane generation now in standardisation keeps PAM4 and doubles the symbol rate again, to around 112 GBd, which puts Nyquist at 56 GHz. Copper at that frequency is unforgiving, and the practical consequence is that the reach of an electrical channel keeps shrinking: a backplane of the kind modelled here is already out of the question, and even the few inches from a switch die to a front-panel cage have become a design problem rather than a piece of routing.

Three responses are visible. The first is more sophisticated receivers: maximum-likelihood sequence estimation, which instead of equalising the channel and then slicing, searches for the most likely transmitted sequence given the known channel response, and which is worth a few dB over a decision-feedback equaliser at the cost of considerable digital power. The second is stronger coding, with concatenated and soft-decision schemes replacing the hard-decision Reed–Solomon codes that have served since 2017. The third, and the most consequential, is to stop sending electrical signals so far: co-packaged optics move the electrical-to-optical conversion into the switch package, and linear-drive pluggable optics remove the retimer from the module so that the host SerDes drives the optical engine directly. Both change the problem from equalising twenty inches of copper to equalising two.

For all of that, the structure of the problem has not changed since 10GBASE-KR. There is a channel you cannot alter, a pulse response that describes it completely, a set of filters whose taps are the solution of a quadratic minimisation, and a budget that decides whether the result works. Everything in this report transfers; only the numbers move.

12 Cheat sheet

Quantity	This channel	What it tells you
Insertion loss at Nyquist	-33.3 dB	The headline difficulty. Above about -30 dB, NRZ needs everything in §5
Cursor, unequalised	0.199	Fraction of the transmitted amplitude arriving on time
Worst-case eye, unequalised	-0.87	Negative means some data patterns invert the decision
Noise enhancement of the receive FFE	-5.59 dB	The price of each linear tap; compare with a DFE tap, which is free
Cursor at the slicer	12.9 mV	Out of 800 mV launched
Total noise	2.15 mV rms	Dominated by crosstalk, not by residual ISI
Q and margin	6.01, -1.36 dB	Short of 10^{-12} ; see §8 for what recovers it
PAM4 penalty	9.5 dB	Structural, not recoverable by equalisation

Working order for a new channel. First, get the S-parameters and check them: passive, causal, reciprocal, correctly de-embedded, and with the mixed-mode port map the right way round. Second, look for structural damage — a via stub notch near Nyquist, a connector resonance, an unterminated branch — because no equaliser fixes a notch. Third, compute the pulse response and read the cursor and the first few pre- and post-cursors; that tells you which blocks you need before you design any of them. Fourth, build the budget and find out what actually dominates. Only then choose the equalisation, and re-check the budget afterwards, because the equaliser you added changed the noise as well as the signal.

Glossary

COM. Channel operating margin: the statistical figure of merit IEEE 802.3 uses to decide whether a channel is compliant, computed from the S-parameters and a reference receiver.

CTLE. Continuous-time linear equaliser. An analogue filter with rising gain, which flattens the channel by attenuating low frequencies.

Cursor. The sample of the pulse response at the chosen sampling phase — the part of a symbol that arrives when it should. Everything else is ISI.

DFF. Decision-feedback equaliser. Subtracts the known contribution of already-decided symbols, so it removes post-cursor ISI without amplifying noise.

FFE. Feed-forward equaliser. A symbol-rate FIR filter, in the transmitter (limited by peak power) or the receiver (limited by noise enhancement).

ICN. Integrated crosstalk noise: near- and far-end crosstalk aggregated into a single rms voltage for the budget.

ISI. Intersymbol interference. Energy from one symbol landing on another, caused by frequency-dependent loss and by reflections.

KP4. The RS(544,514) Reed-Solomon code used by 100G-per-lane Ethernet, which takes a pre-correction error rate of roughly 2×10^{-4} below 10^{-15} .

MLSE. Maximum-likelihood sequence estimation. Searches for the most likely transmitted sequence rather than slicing symbol by symbol.

PAM4. Four-level pulse amplitude modulation: two bits per symbol, three eyes, and a 9.5 dB signal-to-noise penalty relative to NRZ at the same swing.

Pulse response. The waveform produced by a single symbol. Sampled on the UI grid it gives the cursor and the ISI taps, which is everything an equaliser design needs.

Q-factor. The communications Q, named after the Gaussian tail function and defined as $(\mu_1 - \mu_0)/(\sigma_1 + \sigma_0)$ — half the eye opening over the rms noise for a symmetric binary signal. $BER = \frac{1}{2} \operatorname{erfc}(Q/\sqrt{2})$, so $Q = 7.03$ gives 10^{-12} . Unrelated to the resonator quality factor.

Matched-filter bound. $SNR = 2E/N_0$, the most any receiver can extract from a pulse of energy E. A linear equaliser approaches it by combining pulse-response samples coherently and falls short of it by the noise enhancement.

Noise enhancement. The gap between a linear equaliser's coherent signal gain and its incoherent noise gain — the net penalty, not the raw tap norm.

Df, loss tangent. The dielectric dissipation factor. Sets the slope of the channel directly: $\alpha_d \approx 2.3 f \sqrt{Dk} \cdot Df$ dB per inch, with f in GHz.

Dk, dielectric constant. Relative permittivity. Sets propagation delay, characteristic impedance and, through the glass-resin difference, fibre-weave skew.

Via stub. The unused remainder of a plated through-hole. A quarter-wave resonator that puts a deep notch in the channel, removed by back-drilling.

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